

WORKFORCE ADMINISTRATION SOLUTION(DEV)

Date	20-06-2025
Team ID	LTVIP2025TMID29622
Project name	Workforce administration solution(DEV)
Maximum marks	

Chapter – 5

Project Planning & Scheduling

5.1 - Project Planning

Objective

To design and develop a centralized, scalable, and intelligent Workforce Administration Solution that streamlines HR operations such as employee data management, shift scheduling, compliance tracking, and system integrations for mid-to-large enterprises.

📖 Background

HR and operations teams often rely on fragmented tools and manual processes to manage workforce-related tasks. This leads to inefficiencies, data inconsistencies, poor compliance visibility, and increased administrative burden. A modern, unified solution is needed to improve accuracy, reduce manual work, and enable real-time workforce insights.

Goals

- Centralize employee data and workforce records
- Enable dynamic, rules-based scheduling
- Automate compliance tracking and reporting
- Support integration with payroll, time tracking, and HRMS tools
- Improve workforce visibility through dashboards and analytics

Target Users

- HR Managers and Workforce Administrators
- Team Leaders and Department Heads
- Compliance Officers and Operations Managers
- Employees (as self-service users)

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Core Capabilities (Planned MVP)

- Centralized employee profile management
- Smart shift scheduling with availability & role matching
- Real-time compliance alerts and auto-generated reports
- Secure API integrations with third-party systems
- Role-based access control and permissions

Technology Stack (Proposed)

- Frontend: React.js with Material UI / Tailwind
- Backend: Node.js with Express / NestJS
- Database: PostgreSQL (relational)
- Authentication: OAuth2 / JWT-based
- APIs: RESTful & GraphQL
- Hosting: AWS / Vercel / Dockerized containers

Expected Impact

- Reduce administrative time by 30–50%
- Improve workforce data accuracy by 90%
- Minimize compliance-related risks and delays
- Enhance visibility and decision-making through integrated dashboards